

Edge-AI NILM System — Bill of Materials

Hardware component list for the ESP32S3 + PZEM-004T edge sensing node

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#	Component	Spec / Role	Qty	Status
1	Seeed Studio XIAO ESP32S3	Dual-core Xtensa LX7, Wi-Fi/BLE, USB-C — edge compute + dashboard node	1	To source
2	PZEM-004T-100A V3.0 (w/ CT coil)	80–260V AC, 100A, UART/Modbus-RTU — handles voltage, current, and power/PF calculation in one board	1	Confirmed — xcluma listing , Amazon.in , ~Rs. 914
3	230V AC extension board	Host enclosure / mains access point for the CT clamp and L-N tap	1	To source — any basic board works
4	4-pin JST-XH -> Dupont cable	Links the PZEM UART header to the ESP32S3	1	Check if bundled with #2 first; cheap add-on if not
5	Logic-level shifter or 2-resistor divider	Steps the PZEM's 5V TX line down to 3.3V	1	Optional — direct 3.3V wiring often reported to work fine
6	Inline fuse or small circuit breaker	Mains-side overcurrent protection	1	Size to the extension board's rating
7	Insulated terminal blocks / wire nuts + heat-shrink	L/N splice from extension board to PZEM	As needed	—
8	Non-conductive project enclosure	Houses the ESP32S3 + PZEM board together	1	Confirmed necessary — exposed board per listing reviews
9	USB-C cable	Programming + power for the ESP32S3	1	—

Functional Summary

The build reduces to **three functional components**: the **XIAO ESP32S3** (compute and dashboard node), the **PZEM-004T** (all sensing — voltage, current, and the power/PF calculation, in one board), and the **230V extension board** (the mains access point the CT clamps onto and the L/N taps connect to). Items 4–9 above are supporting hardware needed to wire those three together safely — not separate sensing functions.

Tool needed but not itemized: a multimeter, to verify no voltage is present before opening the extension board. Not a permanent part of the build, but required before any mains-side wiring step.